

Simulation 2 Review + Winter Vacation SSPA Special Training Plan

Semester 1 Final Lesson • Review and Reflection • 75 minutes • 1-to-3 Online Lesson

Corresponding textbooks: L19 SSPA Mock Exam (2) Review + Winter Vacation Independent Practice Plan

Core Traps: □ T1 Multi-digit • T2 Decimal • T3 Reverse Questions • T5 Area • T9 Score - Simulation 2 Five Major Score-Losing Areas

SSPA Related: ● **High frequency** Integrated diagnosis of sub-test traps, personalized weakness location

Pre-requisite knowledge: L1–L18 full semester content + L19 mock exam

Objective of this class: ❶ Diagnose the reasons for losing points in the five major traps in Simulation 2 ❷ Self-assessment of personal weaknesses ❸ Develop a 15-minute daily exercise plan during winter vacation ❹ Targeted error correction exercises

Student Name:

Class:

Date:

Time Spent:

I.Mock Exam2 SSPA five major trap disaster areas (L19 review)

Trap 1: T1 multi-digit number - "rounding" approximation trap ● SSPA

- ① When taking approximate values, you must clearly see "which digit to take" - tens of thousands, thousands, hundreds?
- ② Key rules: Look at the target position **The next digit** The number determines whether to "enter" or "round"
- ③ Common mistakes: when you get to the "ten thousand digits", you stop after looking at the thousands digits, without confirming the value of the tens of thousands digits
- ④ Another mistake: forgetting to clear all subsequent digits after placing them (for example, 345,678 rounded to the nearest ten thousand = 350,000, not 35,000)

Example questions T1-1: Similar questions to Question 5 of Simulation 2

Round 2,847,563 to the nearest hundred thousand digits.

✗ Common mistakes

2,800,000

I only looked at 4 (round) in the tens of thousands place, but the hundreds of thousands place is 8. I need to look at the next level in the millions place.

✓ Correct solution

2,800,000

The hundred thousand digit = 8, look at the ten thousand digit = 4 (<5, round down) → 2,800,000

Trap 2: T2 Decimal — Decimal Point Position Trap for Decimal Multiplication ● SSPA

- ① Multiply decimals: first multiply the integers, then count the total number of decimal places and add the decimal point
- ② **The most frequent error |||SEP|||: $0.3 \times 0.4 = 1.2$ (correct = 0.12) - Forgetting that multiplying decimals by decimals will get smaller and smaller**
- ③ The second error: the decimal point is placed in the wrong position - it should be from the far right of |||SEP||| Start counting to the left
- ④ **Verification formula: "Multiplier < 1 → Product < Multiplicand" ($0.3 \times 0.4 \rightarrow$ The answer must be < 0.3)** Start counting to the left
- ④ Verification formula: "Multiplier < 1 → product < multiplicand" ($0.3 \times 0.4 \rightarrow$ the answer must be < 0.3)

Example questions T2-1: Similar questions to Question 2 of Simulation 2

Calculate $0.25 \times 0.4 = ?$

✗ Common mistakes

$0.25 \times 0.4 = 1.00$

Directly $25 \times 4 = 100$, I forgot to count from the right at the decimal point

✓ Correct solution

$0.25 \times 0.4 = 0.100 = 0.1$

$25 \times 4 = 100$, the total number of decimal places is $2 + 1 = 3 \rightarrow 0.100 \rightarrow$ the simplest 0.1

Pitfall 3: T5 Area — Trapezoidal/Polygonal Area Unit Confusion ● SSPA

- ① Area of trapezoid = (upper base + lower base) × height ÷ 2 — **not** (upper base + lower base + height) × 2
- ② Area of parallelogram = base × height (|||SEP||| **not** base × hypotenuse)
- ③ Area of triangle = base × height ÷ 2 — Must divide |||SEP|||
- ③ Area of triangle = base × height ÷ 2 — required **divide by 2** !
- ④ Combined graphics: first divide into basic graphics, find the area respectively, and then add the sum

Example T5-1: Simulation 2 trapezoid area problem

The upper base of the trapezoid is 8 cm, the lower base is 12 cm, and the height is 5 cm. Area = ?

✗ Common mistakes

$$(8+12+5) \times 2 = 50 \text{ cm}^2$$

Confusing the perimeter and area formulas - adding the height and multiplying by 2

✓ Correct solution

$$(8+12) \times 5 \div 2 = 50 \text{ cm}^2$$

(Upper bottom + Lower bottom) × Height ÷ 2 = $20 \times 5 \div 2 = 50$ cm²

Trap 4: T9 fractions - addition and subtraction with different denominators "straight addition and straight subtraction" ● SSPA

- ① Fractions with different denominators **cannot** direct addition and subtraction - they must be divided first
- ② Three steps: find LCM → expand the numerator simultaneously → add the numerators with the same denominator
- ③ The answer must be reduced to **The simplest fraction** |||SEP|||, the improper fraction must be transformed Mixed fractions
- ④ Adding and subtracting mixed fractions: Recommended to convert to improper fractions (least error-prone)
- ④ Adding and subtracting mixed numbers: It is recommended to convert to improper fractions (least error-prone)

Example T9-1: Simulation 2 Fraction Questions

$$\frac{2}{3} + \frac{1}{4} = ?$$

✗ Common mistakes

$$\frac{3}{7}$$

Numerator + numerator, denominator + denominator - different denominators can never be added directly!

✓ Correct solution

$$\frac{11}{12}$$

$$\text{LCM}(3,4)=12 \rightarrow \frac{8}{12} + \frac{3}{12} = \frac{11}{12}$$

Trap 5: T3 inverse question - find the original number when the result is known ● SSPA

- ① The key to the reverse question: start from the final result of |||SEP||| **Auxiliary: Assume the unknown** → Column → **Solve the equation**
- ④ Verification: Substitute the answer back to the original question to check whether it is reasonable **Push back** Operation
- ② When working backwards, addition and subtraction are reciprocal, and multiplication and division are reciprocal: the original "addition" becomes "subtraction" when working backwards.
- ③ Best to use **equation** Auxiliary: Assume unknowns → List → Solve equations
- ④ Verification: Substitute the answer back to the original question to check whether it is reasonable

Example T3-1: Simulation 2 reverse question

Xiao Ming has some candy. After taking |||SEP|||, I bought 15 more pills and now I have 45 pills.

How many grains were there? $\frac{1}{3}$ Later, I bought 15 more pills, and now I have 45 pills. How many grains were there?

✗ Common mistakes

$$45 + 15 = 60, 60 \times \frac{1}{3} = 20$$

Calculate forward, not backward - you bought 15 before reaching 45, you should subtract 15 first

✓ Correct solution

$$(45 - 15) \div (1 - \frac{1}{3}) = 30 \div \frac{2}{3} = 45$$

Working backward: there were 30 pills before buying 15, this is the remaining 23 after eating 13

 **General tips for the five major traps: "Looking at multiple digits, decimal points, memorizing area formulas, dividing fractions first, and working backwards"**

II. Self-Assessment Checklist of Personal Weaknesses (Self-Assessment Checklist)

 Please honestly self-assess the following 10 items and check the ones where you often make mistakes. This is the "bullseye" for your winter vacation training!

✓	Weakness description	Trap category	SSPA frequency
☐	Q1. When approximating multiple digits, the wrong target digit is taken (for example, rounding to the thousands digit is regarded as rounding to the 10,000 digit)	T1 multi-digit	 high
☐	Q2. After approximating multiple digits, forget to clear the following digits to zero.	T1 multi-digit	 high
☐	Q3. Wrong decimal point position in decimal multiplication (especially the 0.3×0.4 category)	T2 decimal	 high
☐	Q4. The decimal point position of the quotient of decimal division is wrong	T2 decimal	 medium
☐	Q5. Forgot to divide the area of the trapezoid by 2, or use the wrong formula (for example, add the height and then multiply)	T5 area	 high
☐	Q6. Area of combined graphics: Forgot to divide or overlapping parts are calculated incorrectly	T5 area	 high
☐	Q7. Direct addition and subtraction of fractions with different denominators (numerator + numerator, denominator + denominator)	T9 score	 high
☐	Q8. I forgot to reduce the fraction answer, or the improper fraction was not converted into a mixed number.	T9 score	 high
☐	Q9. When working backwards in the reverse problem, the direction of operation is reversed (when adding, subtracting, forgetting to change)	T3 reverse	 high
☐	Q10. Missing answer sentences for application questions/Missing units/Confusing columns	T7 Application	 medium

III. Winter SSPA per day 15 minutes special training plan (two weeks = 14 days)

Planning Principles ● SSPA Preparation

- ① Only **15 minutes**—The focus is **continuous** rather than long-term
- ② Focus on **1 trap |||SEP|||**, **practice the five major traps in turn (can be cycled twice in two weeks)** Each exercise: 3 minutes to review the rules → 10 minutes to do the questions → 2 minutes to mark the answers
- ④ Rest on Sunday (or make up for unfinished exercises during the week)
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day	date	focus trap	Exercise content (15 minutes)	✓
1	Day 1	T1 multi-digit	5 questions on taking approximate values + 5 questions on reading and writing numbers	☐
2	Day 2	T2 decimal	5 questions on multiplication of decimals + 5 questions on division of decimals	☐
3	Day 3	T5 area	3 questions on the area of a trapezoid + 3 questions on the area of a triangle + 2 questions on combined figures	☐
4	Day 4	T9 score	5 questions on adding and subtracting different denominators + 5 questions on adding and subtracting mixed fractions	☐
5	Day 5	T3 reverse	5 inverse problems (with equation assistance) + 3 application problems	☐
6	Day 6	T1+T2 mixed	Multi-digit + decimal mixed exercises 10 questions (timed 12 minutes)	☐
7	Day 7	🌿 Rest days (or make up for unfinished questions from Day 1-6)		☐
8	Day 8	T5 area	10 questions on mixed areas of all shapes (timed 12 minutes)	☐
9	Day 9	T9 score	Four mixed fractions 5 questions + 5 application questions	☐
10	Day 10	T3 reverse	5 questions on converse problems + 5 questions on application of equations	☐
11	Day 11	T2 decimal	10 mixed decimal questions (including approximate values)	☐
12	Day 12	comprehensive	Selected 10 questions from the mock paper (limited to 15 minutes)	☐
13	Day 13	Personally the weakest	For the traps with the most incorrect marks on Days 1-12, do 10 more questions.	☐
14	Day 14	📄 Summary: Count the wrong questions in the past two weeks and make a list of "still need to improve"		☐

⚠ Key rules: After each exercise, you must "correct the answer + mark the wrong question + understand the cause of the error." Doing nothing = doing nothing.

IV. L19 Common Mistake for correctness review question (total 20 question · Focus on the five traps)

Group A — T1 more number of digits approximate value (questions 1-4)

📖 Exam tips (SSPA must read)

- ① Do the questions you know first, don't get stuck on one!
- ② Check the unit for each question (cm vs cm² vs cm³)!
- ③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
- ④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
- ⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

#	Question	Difficulty	Working Space
R1	Approximate 3,456,789 to the "hundredthousandth digit".		
R2	Round 67,890,123 to the nearest million.		
R3	Round 4,507,632 to the nearest ten thousand. And write which digit you are looking at.		
R4	Which of the following is a correct approximation to the millionth place of 29,876,543? A. 29,000,000 B. 30,000,000 C. 29,900,000 D. 20,000,000		

Group B — T2 decimalcalculate (questions 5-8)

#	Question	Difficulty	Working Space
R5	Calculate $0.6 \times 0.3 = ?$		
R6	Calculate $0.25 \times 0.8 = ?$		
R7	Calculate $0.15 \times 0.4 = ?$ (Write the correct decimal point position after $15 \times 4 = 60$)		
R8	Calculate $0.05 \times 0.02 = ?$ (Note: How many zeros are there after the decimal point?)		

Group C - T5 areatrap (questions 9-12)

#	Question	Difficulty	Working Space
R9	The upper base of the trapezoid is 6 cm, the lower base is 10 cm, and the height is 4 cm. Area = ?		
R10	Triangular base 8 cm, height 5 cm. Area = ? (Careful: Did you divide by 2?)		
R11	The parallelogram has a base of 9 cm, a height of 6 cm, and a hypotenuse of 7 cm. Area = ?		
R12	A combined figure consists of a trapezoid and a parallelogram. Trapezoid: $(4+8) \times 3 \div 2$, parallelogram: 5×3 . Total area = ?		

Group D - T9 fraction operation (questions 13-16)

#	Question	Difficulty	Working Space
R13	$\frac{1}{3} + \frac{1}{6} = ?$ (Get points first!)		

R14	$\frac{3}{4} - \frac{1}{6} = ?$ (Remember to keep it simple!)		
R15	$1\frac{1}{2} + 2\frac{2}{3} = ?$ (Tip: Convert Improper Fractions)		
R16	$\frac{2}{3} + \frac{1}{4} + \frac{1}{6} = ?$ (Three fractions are combined)		

Group E — T3 reverse towardquestion (questions 17-20)

#	Question	Difficulty	Working Space
R17	Add 15 to a number and then multiply it by 3. The result is 90. Find this number. (Use backwards reasoning)		
R18	After Xiao Ming used the saved SEP , he deposited another \$30, and now he has \$90. How many yuan was it originally? $\frac{1}{4}$ After you use it, you deposit another \$30 and now you have \$90. How many yuan was it originally?		
R19	If you subtract 28 from a number, divide it by 4, and then add 10, the result is 25. Find the original number.		
R20	There is some water in the bucket. After pouring out the SEP , add another 3 liters, now you have 10 liters. How many liters did it originally have? $\frac{2}{5}$ Then, add another 3 liters and now you have 10 liters. How many liters did it originally have?		

V. Additional reinforcement of practice (total 22 questions · full trap mix)

Basic layer (question 21-28 · Everyone must do)

#	Question	Difficulty	Working Space
21	Round 12,345,678 to the nearest million.		
22	Calculate $0.4 \times 0.7 = ?$		
23	Triangular base 10 cm, height 6 cm. Area = ?		
24	$\frac{1}{5} + \frac{3}{10} = ?$		
25	Round 5,678,900 to the nearest hundred thousand digits.		
26	Calculate $0.8 \times 0.05 = ?$		
27	The upper base of the trapezoid is 5 cm, the lower base is 7 cm, and the height is 6 cm. Area = ?		

28	$\frac{2}{3} + \frac{1}{9} = ?$		
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Advanced layer (question 29-36 · Select do)

#	Question	Difficulty	Working Space
29	Take 67,543,210 to the nearest tens of millions and write the Chinese pronunciation.		
30	Calculate $0.35 \times 0.6 = ?$ (Multiply the integers first, then count the decimal places)		
31	The parallelogram has a base of 12 cm and a height of 5 cm. Area = ? (Don't use a bevel!)		
32	$\frac{3}{5} + \frac{2}{3} = ?$ (LCM(5,3)=?)		
33	Three times a number plus 12 equals 48. Find a certain number. (Backward calculation: $48 - 12 = 36$, $36 \div 3 = ?$)		
34	Calculate $0.45 \div 0.5 = ?$ (Hint: divide by 0.5 = multiply by 2)		
35	The upper base of the trapezoid is 3 cm, the lower base is 9 cm, and the height is 8 cm. Area = ?		
36	$2\frac{1}{3} + 1\frac{1}{4} = ?$ (convert improper fractions first)		

challenge layer (question 37-42 · choose do)

#	Question	Difficulty	Working Space
37	If you multiply a number by 0.5 and then add 8, the result is 23. Find the original number.		
38	A rectangle is 15 cm long and 8 cm wide. Cut out a triangle from it (base 8 cm, height 5 cm). Remaining area = ?		
39	$\frac{5}{8} + \frac{3}{4} + \frac{1}{2} = ?$ (Three numbers can be divided into three numbers, LCM(8,4,2)=?)		
40	For a barrel of oil, $\frac{1}{4}$ was used on the first day, and the remaining $\frac{1}{3}$ was used on the second day, leaving 12 liters. How many liters did it originally have? (Double reverse!)		
41	Round 99,876,543 to the nearest million. And explain: why not 100,000,000?		
42	Calculate $0.125 \times 0.8 = ?$ Write out the complete process of "first multiply integers → count decimal places → point decimal points".		

VI. The Lessoncorecommon errorssummary

☐ Identify all trap types in our hall☐ Solve 🌟 basic questions independently (100% correct)☐ Challenge 🌿 Advanced questions (80%+ correct)☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Take approximations to see misalignment	Circle the target position first → look at the next digit → ≥ 5 and < 5 will be rounded → clear the rest
2	Wrong decimal point when multiplying decimals	After multiplying the integers → count the total number of decimal places → click from right to left
3	The area of the trapezoid is forgotten $\div 2$	Remember the formula: (upper base + lower base) $\times$ height $\div 2$, not multiplied by 2
4	Parallelogram uses hypotenuse	Area = base $\times$ height (height is the vertical distance, not the hypotenuse)
5	Direct addition of fractions with different denominators	LCM must be found first through the denominator → the numerator is expanded simultaneously → the denominator remains unchanged
6	The answer is not reduced	Final step: Check if the numerator and denominator have common factors
7	Reverse question inference	Work backwards from the final result: addition, subtraction, multiplication, division, order reversal